Jatindar Kumar Mamnani

BS Computer Science | National University of Sciences and Technology (NUST)
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PROFESSIONAL PROFILE

Skilled and Innovative Computer Vision Researcher with proven expertise in designing and implementing ML models for autonomous systems. Skilled in Python, PyTorch, and TensorFlow, with a strong track record in deep learning and multitask learning projects. Seeking opportunities to further advance machine learning capabilities in real- world applications.

EDUCATION

- BS in Computer Science School of Electrical Engineering and Computer Science , 2.93
- Bachelor of Science in Computer Science National University of Sciences and Technology , (2025)

WORK & INTERNSHIP EXPERIENCE

Vision Research and development company

24-Jun-2024 - 24-Sep-2024

- Conduct research on the applications of computer vision and data science in Autonomous cars
- Design and implement ML models and algorithms
- Assist in collecting and preprocessing relevant datasets
- Contribute to the development and refinement of ML models and algorithms
- Document findings, metho

CodSoft

25-Jul-2024 - 25-Sep-2024

- Communicate with managers to set up campus computers used on campus
- Assess and troubleshoot computer problems brought by students, faculty and staff
- Maintain upkeep of computers, classroom equipment, and 200 printers across campus

PROJECTS

Prompt-No-Code Website Assistant

- Developed an AI-powered no-code chatbot assistant that extracts contextual insights from website URLs for automated responses.
- Engineered an LLM-driven chatbot system using FastAPI, enabling real-time, context-aware user interactions.
- Optimized API-driven chatbot responses for accuracy, automatio

SemEval-Task-8-Table Answering

- Built a tabular QA system using transformer-based NLP for structured data retrieval.
- Implemented text-to-SQL conversion using pre-trained language models to generate structured queries.
- Conducted model evaluation using accuracy, precision, recall, and execution latency benchmarks.
- Addressed NLU

VQA-Based-Medical-Advisor

- Developed a medical visual question-answering system using CNNs for image features and LSTMs/ Transformers for text queries.

- Implemented multimodal fusion to combine image and text features.
- Trained and evaluated on the VQA-Med dataset, using accuracy, BLEU, and METEOR metrics.
- Tackled challen

A Video Based Multi-Task Learning Model for Autonomous Driving

- Developed a cutting-edge multitask learning model for real-time autonomous driving.
- Engineered to perform semantic segmentation, lane detection, and object detection on edge devices with high performance.
- Achieved 82.4% mIoU on the VIL-100 validation set for lane segmentation.
- Demonstrated str

TECHNICAL SKILLS

- Python socket programming
- Cuda
- Python and bash scripting
- Web scraping
- Computer vision
- Data science
- Flask
- Contrcat management

SOFT SKILLS

- Multi tasking
- Group work